

COVER

Industrial Engineering

Course Code: 4024

Program: Diploma in Mechanical Engineering / Tool and Die Engineering / Manufacturing Technology

Semester 4 | Revision 2021 | Program Core | Credits: No Credit

Kerala Polytechnic Study Hub

Professional Diploma Engineering Handbook

COURSE OVERVIEW

Industrial Engineering improves productivity, quality, delivery and cost by studying men, machines, materials, methods and money. It connects PPC, plant layout, maintenance, work study, work measurement, quality control, acceptance sampling, costing and depreciation. Objectives: introduce the role of industrial engineer from production planning to costing; build understanding of PPC, scheduling, forecasting, SQC, TQM, method study and work measurement.

MODULE 1 PPC, PLANT LAYOUT AND MAINTENANCE

PPC functions: forecasting, routing, scheduling, dispatching and follow-up. Forecasting estimates demand. Routing fixes operation path. Scheduling fixes time; Gantt chart shows planned and actual progress. Dispatching releases work orders. Production types: mass, batch and job order. Plant layout types: process, product, fixed position and combination. Material handling equipment: AGV, forklift, crane, conveyor, hoist. Maintenance: breakdown, preventive, predictive, scheduled and condition based. Bathtub curve: early failure, useful life, wear-out.

MODULE 2 WORK STUDY AND WORK MEASUREMENT

Work study is systematic examination of work to improve productivity. Method study records and critically examines existing method to develop better method. Procedure: select, record, examine, develop, define, install, maintain. Tools: operation process chart, flow process chart, two handed chart, man-machine chart, string diagram, flow diagram. Therbligs are basic motion elements; SIMO chart records simultaneous motion. Time study finds observed time, normal time and standard time. Standard time = observed time x rating/100 x (1 + allowance).

MODULE 3 QUALITY CONTROL AND SQC

Quality control ensures product meets specification. Inspection types: first piece, floor and centralized inspection. SQC uses statistical tools to control variation. Normal distribution explains natural variation. Central tendency: mean, median, mode. Dispersion: range, variance and standard deviation. Variables are measured values; attributes are counted qualities. X-bar chart monitors average, R chart monitors range, p chart monitors fraction defective and c chart monitors number of defects.

MODULE 4 ACCEPTANCE SAMPLING, COSTING AND DEPRECIATION

Acceptance sampling inspects a sample to accept or reject a lot. OC curve shows probability of accepting a lot for different quality levels. Producer risk is rejecting a good lot; consumer risk is accepting a bad lot. Sampling plans: single, double and multiple. Estimating predicts probable cost before work. Costing finds actual/calculated cost. Prime cost = direct material + direct labour + direct expenses. Factory cost = prime cost + factory overhead. Selling price = office cost + selling overhead + profit. Depreciation is reduction in asset value due to wear, age, use or obsolescence.

FORMULA BANK

Productivity = Output / Input. Normal Time = Observed Time x Rating / 100. Standard Time = Normal Time x (1 + Allowance percent). Mean = $\sum x / n$. Range = Maximum - Minimum. Variance = $\sum (x - \bar{x})^2 / n$. Standard Deviation = square root of variance. Prime Cost = Direct Material + Direct Labour + Direct Expenses. Factory Cost = Prime Cost + Factory Overhead. Selling Price = Office Cost + Selling Overhead + Profit. Straight Line Depreciation = $(\text{Cost} - \text{Scrap Value}) / \text{Life}$.

SOLVED EXAMPLES

1. Observed time 4.8 min, rating 110 percent, allowance 15 percent. Normal time = $4.8 \times 110/100 = 5.28$ min. Standard time = $5.28 \times 1.15 = 6.07$ min. 2. Measurements 18,20,20,22,25. Mean = $105/5 = 21$. Range = $25 - 18 = 7$. 3. Direct material 4000, direct labour 2500, direct expenses 500, factory overhead 1500, office overhead 1000, selling overhead 500, profit 2000. Prime cost = 7000, factory cost = 8500, office cost = 9500, selling price = 12000.

MODEL QUESTION PAPER

Part A: Define PPC. What is routing? State two objectives of method study. Define standard time. What is SQC? What is OC curve? Define prime cost. What is depreciation? Part B: Explain PPC functions. Compare production types. Explain layout types. Explain method study procedure. Explain $\bar{X}$, R, p and c charts. Part C: Solve standard time problem; solve mean and range problem; explain acceptance sampling risks; calculate selling price; calculate straight line depreciation.

MODEL ANSWER KEY

PPC is planning and control of production from forecasting to dispatching and follow-up. Routing is deciding operation sequence and path. Standard time is normal time plus allowances. SQC is use of statistics for quality control. OC curve shows probability of acceptance. Producer risk: rejecting good lot. Consumer risk: accepting bad lot. Prime cost includes direct material, direct labour and direct expenses. Depreciation is loss in asset value over time. References: M. Mahajan; O.P. Khanna; R. Keith Mobley; NPTEL and SWAYAM resources.